Session ID	Day	Time	Room	Session Title	Capacity	Scheduled Papers	Has Empty Slot	Paper IDs	Titles
S01	Tuesday	09:20 AM	Room A	Best Paper Candidates	3	3	No	184; 211; 785	Coordinating GPU Data Centers and Power Grid Regulation Service for Exogenous Carbon Benefits; FLYING SERVING: On-the-Fly Parallelism Switching for Large Language Model Serving; OCTANE: Breaking the Neighbor-List Bottleneck in GPU Molecular Dynamics
S02	Tuesday	10:50 AM	Room A	Compiler, Code Generation and Autotuning	4	4	No	74; 283; 427; 557	GRASP: Optimizing VLIW Instruction Scheduling via Graph Reinforcement Learning; Continuation-Preserving Tiling for Pointer-Chasing Optimization in Structured Mutual Recursion; S2VEC: Compiler-Driven Stream Specialization for Linearized Vectorization; CKTI: A Domain-Specific Compiler for Lowering CUDA Kernels to Triton-IR
S03	Tuesday	10:50 AM	Room B	Runtime Scheduling and Adaptive Execution	4	4	No	66; 435; 195; 422	Barrier-Aware Task Scheduling for Bulk-Synchronous Parallel Architectures; FaaSlim: Partial Caching of Snapshot-based VMs for Serverless Computing; Block-Aware Adaptive State Management for Optimistic Parallel Discrete Event Simulation; Lock Shielding: A General Technique for Misuse-Resilient Locks
S04	Tuesday	10:50 AM	Room C	Energy & Sustainability	4	4	No	63; 89; 212; 464	Wattchmen: Watching the Wattchers – High Fidelity, Flexible GPU Energy Modeling; Agile QoS-aware Dynamic Power Management with eBPF Governors; SmartCap: Coordinated CPU–GPU Power Capping for Performance-Assurance Energy Efficiency; CATS: Correlation-aware Task Scheduling for GPU Power Optimization in AI Data Centers
S05	Tuesday	01:40 PM	Room B	Performance Modeling & Insight	4	4	No	74; 82; 217; 594	TenProf: A Tensor-Centric Profiler for Deep Learning Workload Analysis and Optimization; GRASP: Fine-grained and Adaptive Sampled Simulation for GPU Performance Modeling; Mantis: Decoding HPC Telemetry Data for Robust System Prediction; ViSim: A Lightweight SpMV Performance Simulator via Statistical and Visual Residual Learning
S06	Tuesday	01:40 PM	Room C	I/O & Storage	4	4	No	12; 177; 285; 446	Harmonia: Enhancing Data Placement and Migration in Hybrid Storage Systems via Multi-Agent Reinforcement Learning; ColdMap: Compaction-Aware Cost-Benefit Zone Cleaning for ZNS-Based Key-Value Stores; CoCache: Accelerating Reads in KV Stores via Cooperative Metadata and Data Cache Management; TOTO: Transparent I/O Tuning for HPC Applications
S07	Wednesday	10:50 AM	Room A	Graph Search & Paths	4	4	No	44; 216; 469; 844	G-PathGen: An Efficient GPU-Parallel k-Critical Path Generation Algorithm; Parallel Bidirectional A* Search for GPU-Accelerated Pathfinding; MPMOS: Massively Parallel Multi-Objective Shortest Paths; Distributed Disjoint Weighted Matchings in Demand-Aware Reconfigurable Optical Datacenters
S08	Wednesday	10:50 AM	Room B	AI Training Systems	4	4	No	68; 131; 282; 392	Closing the Efficiency Gap: AI Datacenter Co-design Roadmap for Scalable Training of LLMs; COMETS: Cost-effective Multi-node Efficient Training System with Memory Pooling and Sharing; SPPO: Making Million-Token LLM Training Practical on Modest GPU Clusters; Rudder: Steering Prefetching in Distributed GNN Training using LLM Agents
S09	Wednesday	10:50 AM	Room C	Communication & Collectives	4	4	No	19; 187; 387; 78	SHIRO: Near-Optimal Communication Strategies for Distributed Sparse Matrix Multiplication; PACER: A Userspace Network Rate Controller in MPI with Adaptive Compression for Parallel Applications; Scalable All-to-allv Algorithms for Dynamic and Irregular Communication Patterns; StencilMD: Optimizing Communication in Molecular Dynamics Simulations
S10	Wednesday	01:40 PM	Room A	Graph Analytics	4	4	No	47; 160; 193; 344	DCSM: Enabling Inter-Batch Parallelism for Continuous Subgraph Matching on GPU; SVSIG: Incremental Streaming Graph Processing with Source Vertex Suppression; GAAF: Fast and Scalable Graph-based Vector Similarity Search with Any-Match Label Filtering; HoloGraph: Bridging the Throughput Gap in Heterogeneous Graph Pattern Matching via Workload-Aware Steering
S11	Wednesday	01:40 PM	Room B	LLM Serving	4	4	No	155; 222; 231; 268	Taming Dynamic Diffusion LLM Inference through Virtual Static Execution; SparseServe: Unlocking Parallelism for Dynamic Sparse Attention in Long-Context LLM Serving; InferFast: Bridging the Gap Between Unstructured LLM Sparsity and Practical GPU Throughput; dLLM-Serve: Bridging the Memory Gap in Diffusion Language Model Serving
S12	Wednesday	01:40 PM	Room C	Near-Memory Architectures	4	4	No	38; 190; 272; 521	DANMP: Accelerating Multi-Scale Deformable Attention Using Near-Memory-Processing Architecture; RPFC: A Router Partitioning and Forward Channel Routing Framework for 2.5D MCM System; MyT: Efficient Manycore based on Many Threading and Scalable Memory Parallelism; Clutch: High Performance Vector-Scalar Comparison using DRAM via Chunked Temporal Coding
S13	Wednesday	03:30 PM	Room A	Graph Traversal & Connectivity	5	5	No	225; 243; 485; 668; 604;	cuMIS: A Unified Scalable Framework for Computing Maximal Independent Sets on Trillion-Edge Graphs; BLEST: Blazingly Efficient BFS using Tensor Cores; CORE-BFS: Communication Optimized Rectangular Partitioned BFS Achieving 160.845 TeraTEPS on Frontier Supercomputer;DynLP: Parallel Dynamic Batch Update for Label Propagation in Semi-Supervised Learning; Asynchronous Distributed Algorithms for Maximal-Weighted Graph Matching
S14	Wednesday	03:30 PM	Room B	Data Analytics & Compression	5	5	No	186; 593; 425; 639; 32	GPZ: GPU-Accelerated Lossy Compressor for Particle Data; GFAz: State-of-the-Art Graphical Fragment Assembly Compression; Optimizing Streaming Tensor Decomposition on GPU; DA-MLAD: Drift-Decomposed Meta-Learning for Continual Log Anomaly Detection in Supercomputing Systems; TADS: Trend-Aware Dynamic Load Balancing for Large-Scale SNN Simulations with Delay-Sharded Graph Infrastructure

S15	Wednesday	03:30 PM	Room C	CXL & Memory Systems	5	5	No	134; 159; 163; 171; 784	CXL-CCL: Inter-Node Collective GPU-Communication Using a CXL Shared Memory Pool; IBEX: Internal Bandwidth-Efficient Compression Architecture for Scalable CXL Memory Expansion; Clone: A Collaborative Multi-device System for Retrieval-Augmented Generation over CXL; Anchoring Whole-System Persistence and Resilience in CXL; Griffin: Coherency-Aware Task Scheduling and Memory Allocation for CXL Interconnects
S16	Thursday	09:20 AM	Room A	Cross-Layer Performance Optimization	3	3	No	151; 147; 90	THAC: Unlocking Performance in Parallel HPC Applications via UQ-Aware Automated Approximation; Cross-Architecture Autotuning for Single-Source Heterogeneous Programming Models; Look Before You Leap : Precision Instruction Supply via SmartScout
S17	Thursday	10:50 AM	Room A	GPU-Accelerated Query and Geometry	4	3	Yes	555; 27; 53	Parallel Query Processing through Optimal Key Grouping on GPU-Based B+-Trees; X-HD: Fast Hausdorff Distance Computation with Ray Tracing; Rethinking Collision Detection on GPU Ray Tracing Architecture
S18	Thursday	10:50 AM	Room B	AI Inference	4	4	No	23; 281; 292; 650	LayerScope: Predictive Cross-Layer Scheduling for Efficient Multi-Batch MoE Inference on Legacy Servers; Aurora: A Disaggregated GPU-PNM-PIM System for High-Throughput Mixed-Length LLM Inference; HOPO: Accelerating Multimodal Neural Networks Inference via Holistic Parallelism Optimization; Latency-SLO-Aware Memory Offloading for Large Language Model Inference
S19	Thursday	10:50 AM	Room C	Resilience and Error Detection	4	4	No	56; 366; 388; 610	Not All Errors Are Equal: A Systematic Study of Error Propagation in Large Language Model Inference; SpinTune: Improving the Reliability of Quantum Sensor Networks for Practical Quantum-Classical Utility; Harnessing MPI mutations for AI error detection; StreamGuard: Low-Overhead Resilience for Real-time HPC Data Streams
S20	Thursday	01:40 PM	Room A	Sparse & Tensor Kernels	4	4	No	25; 415; 501; 277	Parametric Mappings for Distributed Tensor Computations; Communication-Avoiding SpGEMM via
S21	Thursday	01:40 PM	Room B	AI Kernels & Parallelism	4	4	No	269; 381; 475; 766	HPMD: Enabling Hybrid Parallelism with Multi-Dimensional Adaptive DNN Training; EPLoN:
S22	Thursday	01:40 PM	Room C	Energy-Aware Systems	4	4	No	144; 278; 584; 307	DEFT: Joint Task Placement and DVFS for Energy-Efficient Multi-GPU Runtimes; Phase-aware
S23	Thursday	03:30 PM	Room A	Numerical & Scientific Kernels	5	5	No	159; 252; 612; 679; 87	WindStencil: Unleashing GPU Potential for High-Order Stencil Computation in High-Performance
S24	Thursday	03:30 PM	Room B	Efficient Privacy Computing	5	5	No	95; 107; 117; 302; 685	MegaZK: A Memory Efficient GPU System Accelerating End-to-end Zero-Knowledge Proof;
S25	Thursday	03:30 PM	Room C	Quantum Computing	5	5	No	48; 67; 375; 424; 524	quEStab: Towards Scalable Quantum Circuit Simulation on Multi-GPU using an Extended Stabilizer